

Beyond the marginal distribution: long-range correlations in prime gaps imprint on the 3D conformation of a self-avoiding polymer

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Version 3.0 (consistency, protocol-dependence, and reporting revision)

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Abstract

The folding of a heteropolymer is governed by its monomer sequence, yet the intermediate regime between periodic and random sequences—those that are aperiodic but deterministic—remains largely unexplored. We introduce the Prime-Driven Topological Polymer (PDTP), a coarse-grained self-avoiding chain in which the local bending stiffness is set by an exponential map of the local prime gap, so that dense prime clusters act as flexible hinges and the long inter-prime runs as rigid rods. Using overdamped Langevin dynamics with simulated annealing, we ask whether the folded state is fixed by the composition of this sequence or by its arithmetic ordering. To separate the two, we compare the true prime chain against surrogates that preserve the marginal stiffness distribution exactly while destroying sequential order. The gross conformation is surrogate-invariant: neither the radius of gyration nor the tail-weighted core-shell amplitude distinguishes the true chain from a shuffle. The finer organization does: the Spearman correlation between stiffness and radial position is significantly suppressed for the true sequence ($\rho = +0.105$ vs. $+0.137$ at $N = 4 \times 10^4$; $p = 6.5 \times 10^{-5}$, $d = -0.96$), an effect a decomposition attributes to ordering rather than to the gap distribution. The imprint is quantitatively modest ($\sim 1\%$ of the radial-rank variance) but statistically robust and reproducible across system sizes, demonstrating that deterministic arithmetic order leaves a measurable trace on three-dimensional structure—one that survives the entropically favored radial sorting of the collapse. Scale-selective surrogate tests (a block shuffle and its converse, a local scramble) further localize this order: the imprint is destroyed by global rearrangement yet wholly preserved under complete *local* scrambling within windows of up to $\sim 10^2$ consecutive gaps, revealing that the driver is an irreducible, *macroscopic* long-range arithmetic correlation of length $\xi \approx 3.5 \times 10^2$ gaps ($\approx 30\%$ of the sequence, under the reference annealing protocol; ξ is a protocol-conditioned observable of the annealed ensemble) and falsifying the local-cluster (*topological-trap*) hypothesis of version 1.

Note (Version 2.0). This revision supersedes version 1.0. The mechanism proposed there—that local twin-prime clusters act as *topological traps*—has since been tested and *falsified* by two new scale-selective surrogate experiments (the S4 block shuffle and the S5 local scramble), which instead localize the ordering to a macroscopic correlation length $\xi \approx 3.5 \times 10^2$ gaps (Sec. 5.2). The abstract, Sec. 5.2, and the Conclusion are updated accordingly; all version 1.0 data and results stand unchanged.

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Note (Version 3.0). This is a reporting and consistency revision: no version 2.0 datum is altered, and no new production ensemble is introduced. The changes are: (i) the error-bar convention (mean $\pm$ s.e.m.) is now stated explicitly; (ii) the annealing depth of every ensemble is recorded in the corresponding figure caption, in absolute steps and in steps per bead, resolving the apparent discrepancy between the $N = 10^4$ points of Figs. 3 and 4; (iii) the correlation length ξ of Eq. (8) is now explicitly identified as a *protocol-conditioned* observable of the annealed ensemble, and a note added in proof reports that a deeper-annealed re-measurement in follow-up work shifts its value upward at fixed N while leaving the macroscopic-order conclusion intact (Sec. 5.2); (iv) the construction-level coupling between segment length and segment stiffness, and the treatment of the chain-end boundary segments, are now stated explicitly in Sec. 2; (v) the reduced statistical power of the R_g comparison ($n = 10$) is flagged (Sec. 4.1); (vi) the code and data are now archived with a persistent DOI (Data availability); and (vii) the Introduction and Sec. 5.3 now engage prior art that version 2.0 omitted—correlated-sequence random heteropolymers, the conformation-dependent (“protein-like”) sequence-design programme and its Lévy-flight sequence correlations, Fibonacci-sequence molecular chains, the proposal of polymeric compaction as a probe of sequence correlations, and the effective limit-periodicity/hyperuniformity of the primes—and position the present contribution relative to each.

1 Introduction

The folding of a heteropolymer into a compact, organized state is a canonical problem of soft-matter physics, underlying phenomena from protein structure to chromatin architecture [1, 2]. A primary determinant of the emergent morphology is the monomer sequence—the one-dimensional arrangement of chemically or mechanically distinct units along the backbone [3]. Two limiting classes of sequence have been studied in depth. At one extreme lie deterministic *periodic* sequences, epitomized by block copolymers, whose translational symmetry drives microphase separation into lamellar, cylindrical, and related ordered mesophases [4, 5]. At the other extreme lie *random* heteropolymers, in which the sequence is a quenched realization of uncorrelated disorder and the folding landscape is governed by frustration and glassy statistics [6, 7]. Between these poles lies a comparatively unexplored regime: sequences that are neither periodic nor random, but *aperiodic and deterministic*, carrying long-range correlations of purely structural origin. This intermediate regime is not empty: analytic work on random heteropolymers with power-law-correlated sequences indicates that long-range sequence correlations qualitatively reshape the folding landscape [28], the conformation-dependent sequence-design programme showed that globule-derived “protein-like” sequences carry Lévy-flight-type long-range correlations coupled to core-shell organization [26, 27], and deterministic aperiodic (Fibonacci) sequences have been mapped onto molecular chains and their conformational scaling examined [29]. What has been lacking is a *controlled attribution*: a demonstration, against nulls that preserve the marginal composition exactly, that the sequential *ordering* of a deterministic aperiodic sequence—as opposed to its composition—leaves a measurable imprint on the three-dimensional conformation of a collapsed chain. That is the gap this paper addresses.

Number theory has repeatedly furnished physics with precisely this kind of structured aperiodicity. The statistics of the nontrivial zeros of the Riemann zeta function coincide with the spectral fluctuations of random-matrix ensembles, forming a celebrated bridge between arithmetic and quantum spectra [8, 9]. In real space, aperiodic tilings and quasicrystals demonstrate that deterministic, non-periodic order can generate sharp diffraction and hierarchical organization absent from both crystals and glasses [10, 11]. The sequence of prime numbers occupies a distinctive place in this

landscape: it is fully deterministic yet globally aperiodic, its coarse density decays only logarithmically as dictated by the prime number theorem, and its local structure—encoded in the sequence of prime gaps g_i , the distances between consecutive primes—carries well-documented correlations that are neither periodic nor memoryless [12, 13]. Indeed, viewed as a one-dimensional point pattern, the primes in suitable large intervals were recently shown to be *effectively limit-periodic* and hyperuniform, exhibiting genuine multiscale order in their structure factor [24, 25]—independent evidence that the prime sequence carries macroscopic order of the kind a physical system might register. These properties make the prime-gap sequence a natural candidate for a designed, deterministic constraint to be mapped onto physical space, in the same spirit that quasiperiodic order has been imprinted onto photonic and mechanical metamaterials [14].

We realize this mapping mechanically. In the Prime-Driven Topological Polymer (PDTP), each bead i of a coarse-grained bead–spring chain is assigned a local bending stiffness that is an exponential function of its prime gap [Eq. (1)]. Dense clusters of primes (short gaps) become flexible hinges, while the rare long gaps become nearly rigid rods. Upon collapse, this construction establishes a competition between two tendencies. The first is generic to any stiff–flexible copolymer: excluded-volume repulsion and bending energy tend to expel rigid segments toward the periphery while flexible segments accommodate the curvature of the dense core, yielding a positive correlation between local stiffness and radial position [15, 16]. The second is specific to the arithmetic origin of the sequence: the deterministic ordering of the gaps along the backbone constrains which segments can reach the surface. The central question is whether this arithmetic ordering leaves any measurable imprint on the folded state, or whether the conformation is fixed entirely by the *marginal* distribution of stiffnesses, indifferent to their sequence. To resolve this we introduce a surrogate-based null model that preserves the marginal distribution exactly while destroying sequential order, and show—through a large ensemble of annealed trajectories—that the arithmetic ordering does leave a small but statistically robust signature, invisible in gross morphology and in marginal-matched nulls, yet detectable in the rank organization of the collapse.

2 Model definition

2.1 Prime skeleton and gap extraction

We identify the beads of a chain of length N with the integers $i = 0, 1, \dots, N - 1$. A sieve of Eratosthenes partitions these indices into primes and composites [12]. For a composite bead lying in the maximal run of consecutive composites bounded by the primes $p_{\text{left}} < i < p_{\text{right}}$ we define a *local prime gap* $g_i = p_{\text{right}} - p_{\text{left}}$, assigning every bead within a given inter-prime run the same value; for a prime bead $g_i = 0$. The boundary segments take $g_i = p_1$ for $i < p_1$ and $g_i = N - p_{\text{last}}$ for $i > p_{\text{last}}$; these end conventions affect only a handful of beads (fewer than one gap length at either terminus, a fraction $\lesssim 10^{-3}$ of the chain) and are immaterial to the rank statistics reported below. Within our largest system, $N = 4 \times 10^4$, the extremal gap is $g_{\text{max}} = 72$, between the primes 31397 and 31469.

We note a construction-level coupling that is a deliberate design choice: because every bead of an inter-prime run inherits the same gap value, a long gap produces a segment that is simultaneously *longer* (its contour length is proportional to g_i) and *stiffer* (its modulus grows as $e^{\alpha g_i}$, Sec. 2.2). The two amplifications compound, which is precisely what turns the rare long gaps into mechanically distinctive rigid rods. Every surrogate used in this paper inherits the identical coupling from its own anchor pattern, so all true-versus-surrogate contrasts compare like with like; the coupling shapes the *absolute* morphology, not the controlled differences on which our conclusions rest.

2.2 Gap-driven exponential stiffness

The gap field is mapped onto the local bending stiffness k_i through a saturated exponential map,

$$k_i = \begin{cases} k_{\text{hinge}}, & i \text{ prime}, \\ \min[k_{\text{hinge}} e^{\alpha g_i}, k_{\text{max}}], & i \text{ composite}. \end{cases} \quad (1)$$

The exponential rather than linear map is essential: because the gap envelope varies only slowly, a linear coupling produces a stiffness field too smooth to break the conformational symmetry, whereas the exponential map converts modest gap fluctuations into a stiffness contrast spanning nearly four orders of magnitude ($k_{\text{max}}/k_{\text{hinge}} = 2000$ at our parameters). The amplification coefficient α and cap k_{max} are fixed by an independent stability calibration (Sec. 3).

2.3 Coarse-grained potential energy

Bonds between consecutive beads are harmonic,

$$E_{\text{bond}} = \frac{1}{2} k_{\text{bond}} \sum_{i=1}^{N-1} (|\mathbf{r}_i - \mathbf{r}_{i-1}| - b_0)^2, \quad (2)$$

with rest length b_0 . The gap-driven stiffness enters through a cosine bending potential on interior triplets,

$$E_{\text{bend}} = \sum_{i=1}^{N-2} k_i (1 - \cos \theta_i), \quad \cos \theta_i = \frac{(\mathbf{r}_i - \mathbf{r}_{i-1}) \cdot (\mathbf{r}_{i+1} - \mathbf{r}_i)}{|\mathbf{r}_i - \mathbf{r}_{i-1}| |\mathbf{r}_{i+1} - \mathbf{r}_i|}, \quad (3)$$

so that k_i acts as a local, sequence-dependent bending modulus [15, 16]. Self-avoidance is a purely repulsive Weeks–Chandler–Andersen (WCA) interaction between non-bonded pairs ($|i-j| > 1$) [17],

$$E_{\text{WCA}}(r) = \begin{cases} 4\varepsilon [(\sigma/r)^{12} - (\sigma/r)^6] + \varepsilon, & r < r_c = 2^{1/6}\sigma, \\ 0, & r \geq r_c, \end{cases} \quad (4)$$

evaluated with a k -d-tree neighbor list and with the pairwise force magnitude capped at $f_{\text{cap}}^{\text{WCA}}$ as a numerical safeguard. Finally, a weak radial restoring force tethers each bead to the instantaneous center of mass $\mathbf{r}_{\text{cm}}$,

$$\mathbf{F}_i^{\text{conf}} = -k_{\text{conf}} (\mathbf{r}_i - \mathbf{r}_{\text{cm}}), \quad (5)$$

with k_{conf} small: it gathers the globule without crushing the rigid rods, allowing the gap-driven gradient to survive collapse.

2.4 Overdamped Langevin dynamics and annealing

The chain is propagated with overdamped (Brownian) Langevin dynamics in reduced units ($\sigma = \varepsilon = k_B = 1$) [18]:

$$\mathbf{r}_i \leftarrow \mathbf{r}_i + \mu \mathbf{F}_i \Delta t + \sqrt{2\mu k_B T \Delta t} \boldsymbol{\xi}_i, \quad \boldsymbol{\xi}_i \sim \mathcal{N}(\mathbf{0}, \mathbb{I}_3), \quad (6)$$

where $\mathbf{F}_i = -\nabla_i E$, μ is the mobility, and $\boldsymbol{\xi}_i$ is independent Gaussian noise. Compaction is achieved by simulated annealing [19], lowering the temperature geometrically,

$$k_B T(t) = k_B T_{\text{hi}} (T_{\text{lo}}/T_{\text{hi}})^{t/(n_{\text{steps}}-1)}. \quad (7)$$

Per-bead displacements are capped at $|\Delta \mathbf{r}_i| \leq \delta_{\text{max}}$, and each trajectory is initialized as a correlated random walk (persistence 0.55). Parameters are listed in Table 1.

Table 1: Model and protocol parameters (reduced units).

Symbol	Value	Meaning
b_0	1.0	bond rest length
k_{bond}	45.0	bond stiffness
k_{hinge}	0.25	baseline (prime) bending stiffness
α	0.15	gap amplification coefficient (calibrated)
k_{max}	500.0	bending-stiffness cap (calibrated)
σ, ε	1.0, 1.0	WCA diameter, energy scale
r_c	$2^{1/6}\sigma$	WCA cutoff
$f_{\text{cap}}^{\text{WCA}}$	30.0	WCA force cap
k_{conf}	0.02	weak confinement constant
μ	1.0	mobility
Δt	0.008	integration step
δ_{max}	0.25	per-bead displacement cap
$k_B T_{\text{hi}}, k_B T_{\text{lo}}$	0.50, 0.003	annealing temperature range

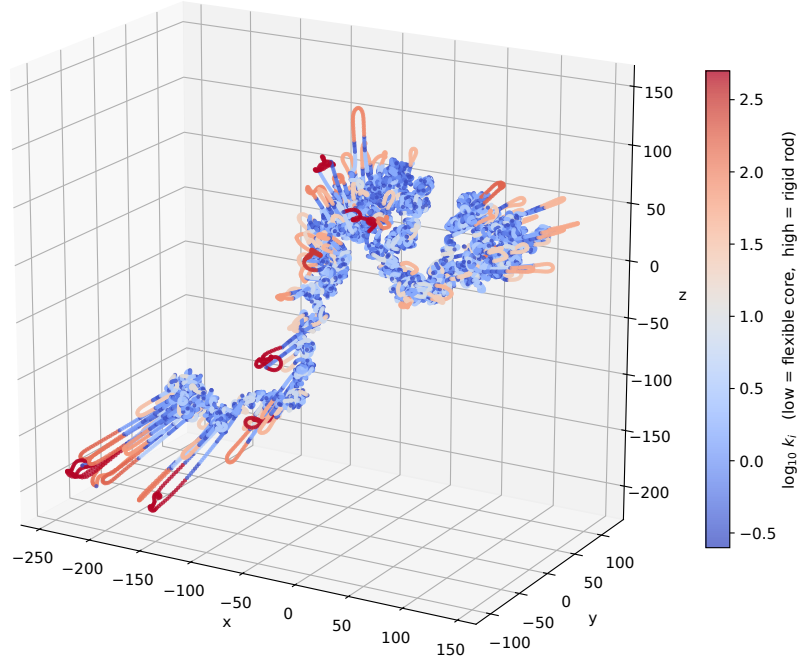


Figure 1: Representative folded conformation of the PDTP chain at $N = 4 \times 10^4$ (overdamped Langevin with simulated annealing, 3800 steps, seed 42). Beads are colored by $\log_{10}$ of the local bending stiffness k_i [Eq. (1)], from flexible prime-rich hinges (blue, $k_i = k_{\text{hinge}} = 0.25$) to rigid long-gap rods (red, capped at $k_{\text{max}} = 500$). The chain collapses to a compact globule ($R_g = 110.7$) in which dense flexible cores are threaded by rigid rods extending into peripheral loops; for this conformation the stiffness–radius rank correlation is $\rho = +0.065$ and the tail-weighted core–shell separation $\Delta R = +21.0$.

3 Pre-flight calibration

Because the stiffness field of Eq. (1) is amplified exponentially, the rare long-gap sites carry bending moduli orders of magnitude above the baseline, and an explicit integrator can in principle diverge within a single step. We bound the parameters *a priori* through a worst-case stress test on an isolated segment [20]. We locate the global maximum gap ($g_{\max} = 72$, between the primes 31397 and 31469) and extract the 150-bead window centred on it (indices [31358, 31508]). The segment is evolved under bond and gap-driven bending forces alone, with the production displacement cap *removed* so that any latent divergence is exposed, while a small thermal noise ($k_B T = 0.05$) continually injects curvature.

Two findings emerge. First, the cosine bending potential of Eq. (3) is *self-limiting*: its restoring force scales as $\sin \theta_i$ and vanishes as a segment straightens, so a rigid rod relaxes toward the straight configuration without overshoot. Sweeping $\alpha \in [0.05, 0.15]$ —and, independently, relaxing the cap to the fully uncapped limit, where the stiffness at $g_{\max}$ reaches $k \approx 1.2 \times 10^4$ (nearly $5 \times 10^4 k_{\text{hinge}}$)—produces no divergence over the test window. The exponential amplification is thus *not* the binding stability constraint. Second, the binding constraint is set by the stiffest *harmonic* term, the bonds: sweeping the timestep, the scheme is stable for $\Delta t \leq 0.01$ but diverges within a few steps for $\Delta t \gtrsim 0.02$, consistent with the fastest bond-mode limit $\Delta t \lesssim 2/(4\mu k_{\text{bond}}) \approx 0.011$. Guided by this, the production runs adopt $\alpha = 0.15$ with a conservative cap $k_{\max} = 500$ (a factor ≈ 25 below the uncapped worst case) and a hard displacement bound $\delta_{\max} = 0.25$; with $\Delta t = 0.008$ lying a factor ≈ 2.5 below the divergence threshold, all trajectories operate well inside the stable regime. The calibration’s purpose is not to maximize the stiffness contrast but to certify that the symmetry-breaking map can be integrated without numerical artifacts that could masquerade as structural signal.

4 Results

We compare the true prime chain against surrogates under identical force fields, dynamics, and annealing; the only difference between arms is the anchor pattern generating the stiffness field. Three observables characterize the collapsed state: the radius of gyration R_g ; the percentile core-shell separation ΔR , the mean radial distance of the stiffest 20% of beads minus that of the most flexible 20%; and the Spearman rank correlation ρ between local stiffness k_i and radial distance. Positive ρ or ΔR indicates rigid segments at the periphery. Ensembles comprise n independent trajectories per arm. Throughout the paper, values quoted as $x \pm y$ denote the ensemble mean $\pm$ the standard error of the mean (s.e.m.), and box plots show medians and quartiles; p -values are two-sided Welch t -tests unless a Mann–Whitney U test is stated, and d is Cohen’s d .

4.1 Global collapse is surrogate-invariant

All arms collapse along indistinguishable trajectories and their gross dimensions do not separate. At $N = 10^4$ the radius of gyration is $R_g = 57 \pm 16$ (true) versus 49 ± 13 (shuffle) (Welch $p = 0.20$, $n = 10$); we note that this comparison derives from the initial low-statistics sweep and, at $n = 10$ per arm, resolves only large differences—it bounds, rather than excludes, any R_g effect. The gross core-shell amplitude, measured at full statistics, is likewise unresolved: $\Delta R = 14.4 \pm 0.7$ versus 15.1 ± 0.5 at $N = 4 \times 10^4$ ($p = 0.37$, $n = 40$). Within our sensitivity, the overall size and the gross magnitude of stiffness segregation are therefore fixed by the marginal stiffness distribution, which every surrogate preserves.

4.2 A core-shell ordering signature

The rank correlation ρ reveals a signature of ordering. In our decisive experiment— $N = 4 \times 10^4$, deep annealing (8000 steps, i.e. 0.20 steps/bead, matched to the per-bead depth of the $N = 10^4$ reference protocol), $n = 40$ per arm—both arms exhibit positive ρ , but the sorting is significantly *weaker* for the true chain: $\rho = +0.105 \pm 0.007$ versus $+0.137 \pm 0.004$ (Welch $p = 6.5 \times 10^{-5}$, Mann-Whitney $p = 3.1 \times 10^{-4}$, $d = -0.96$; Fig. 2). The deterministic ordering thus imposes a reproducible *resistance* to the radial sorting generic stiff-flexible physics would otherwise produce. The effect is quantitatively modest—stiffness accounts for only $\rho^2 \approx 1\%$ of the radial-rank variance—but statistically robust. The same comparison in ΔR is not resolved at this size ($p = 0.37$): the ordering signal resides in the global rank structure, not in the stiffness extremes, which is why the low-variance estimator ρ is required.

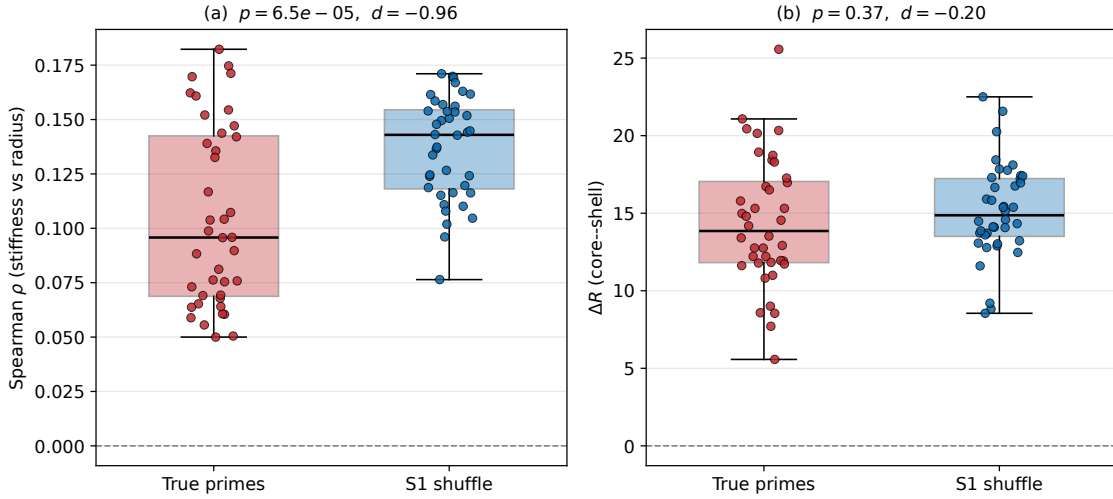


Figure 2: Decisive test at $N = 4 \times 10^4$, deep annealing (8000 steps, 0.20 steps/bead), $n = 40$ per arm; values are mean $\pm$ s.e.m. (a) Spearman ρ : true primes $+0.105 \pm 0.007$ vs. shuffle S1 $+0.137 \pm 0.004$ (Welch $p = 6.5 \times 10^{-5}$, Mann-Whitney $p = 3.1 \times 10^{-4}$, $d = -0.96$)—the true sequence is significantly less radially sorted. (b) $\Delta R = 14.4 \pm 0.7$ vs. 15.1 ± 0.5 ($p = 0.37$, $d = -0.20$), statistically indistinguishable. Boxes: median and quartiles; points: individual runs.

4.3 Ordering versus marginal distribution

To attribute the effect we compare three arms at $N = 10^4$, $n = 40$: the true chain; the Shuffle (S1), preserving the exact gap multiset but randomizing order; and a Cramér surrogate (S3), placing pseudo-primes at density $1/\ln i$, preserving only the coarse density trend [13]. The decomposition is clean (Fig. 3). The ordering contrast (true vs. S1) is strongly significant ($\Delta\rho = -0.051$, $p = 1.6 \times 10^{-5}$, $d = -1.05$). The marginal-shape contrast (S1 vs. S3), both order-randomized, is small and not significant ($\Delta\rho = -0.017$, $p = 0.09$, $d = -0.38$). The combined contrast (true vs. S3) is largest ($\Delta\rho = -0.068$, $p = 1.2 \times 10^{-6}$, $d = -1.18$). Arithmetic ordering therefore accounts for the bulk of the effect ($\approx 75\%$ of the combined shift), while the shape of the gap distribution contributes little. This is the central result: what the folded conformation registers is not the composition of the prime-gap sequence but its sequential, long-range arithmetic structure.

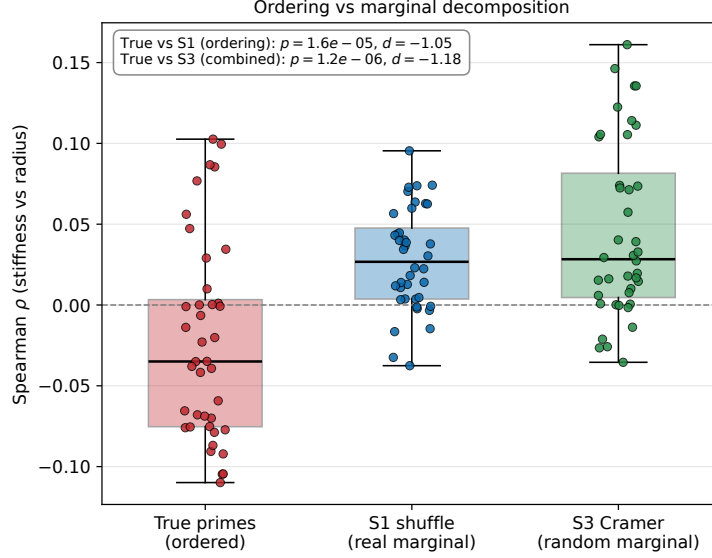


Figure 3: Ordering versus marginal-distribution decomposition at $N = 10^4$, $n = 40$ per arm, reference annealing protocol (2000 steps, 0.20 steps/bead); values are mean $\pm$ s.e.m. True primes ($\rho = -0.023 \pm 0.010$), shuffle S1 ($+0.028 \pm 0.005$; exact gap multiset, order randomized), and Cramér surrogate S3 ($+0.045 \pm 0.009$; pseudo-primes at density $1/\ln i$). Ordering (true vs. S1): $p = 1.6 \times 10^{-5}$, $d = -1.05$. Marginal shape (S1 vs. S3): $p = 0.09$ (n.s.). Combined (true vs. S3): $p = 1.2 \times 10^{-6}$, $d = -1.18$.

4.4 Finite-size behaviour

Finally, we examine the size dependence across $N = 10^4, 2 \times 10^4, 4 \times 10^4$ (Fig. 4). In both ρ and ΔR , the true chain lies systematically below the shuffle at every size, with significance at $N = 10^4$ and 2×10^4 (ρ : $p = 3 \times 10^{-3}$ and 2×10^{-3}). An initial sweep at $N = 4 \times 10^4$ with reduced statistics ($n = 20$) and shallower annealing returned a gap below threshold ($p = 0.075$); the decisive experiment of Sec. 4.2 shows this to be a power-and-protocol artifact rather than a genuine fade—at matched per-bead annealing depth and $n = 40$, the same gap is recovered at $p = 6.5 \times 10^{-5}$. We caution that the *absolute* correlation grows with N for both arms (from $\rho \approx 0$ at $N = 10^4$ to $\rho \approx +0.1$ at $N = 4 \times 10^4$), reflecting both progressive collapse and the varying annealing depth of the sweep; consequently only the fixed- N gap between true and shuffle constitutes a controlled comparison, and we read the absolute N -dependence as a trend rather than an equilibrium scaling law. Within the range studied, once statistics and annealing depth are controlled, the ordering signature is present and significant at every size.

5 Discussion

5.1 What the fold registers

The representative conformation (Fig. 1; $R_g \approx 111$) displays the generic architecture of a semiflexible heteropolymer: dense flexible cores threaded and flanked by rigid rods that extend into loops toward the periphery. Both the true chain and its surrogates share this gross morphology; their difference lies elsewhere. Neither R_g nor ΔR distinguishes the true sequence from a stiffness-matched shuffle;

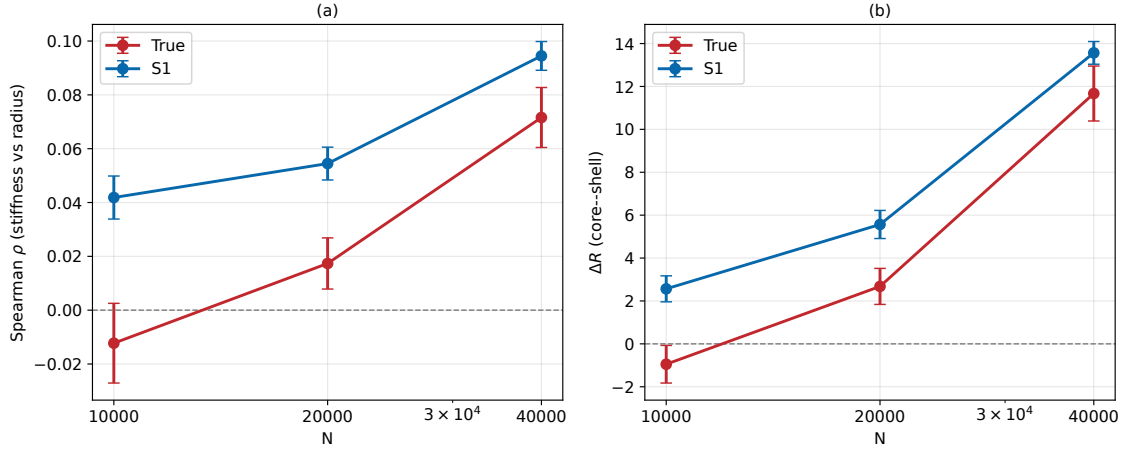


Figure 4: Finite-size behaviour, true vs. shuffle S1, for $N = 10^4, 2 \times 10^4, 4 \times 10^4$ ($n = 20$ per arm; annealing 2000/3500/5000 steps, i.e. a per-bead depth of 0.20/0.175/0.125 steps/bead that *decreases* with N —this sweep is therefore not depth-matched across sizes, and its $N = 10^4$ ensemble ($n = 20$) is distinct from the $n = 40$ ensemble of Fig. 3, which is why the absolute values differ slightly between the two figures). (a) Spearman ρ and (b) ΔR vs. N (mean $\pm$ s.e.m.). The true chain lies below the shuffle at every size. The gap is significant at $N = 10^4$ and 2×10^4 ; the reduced-statistics point at $N = 4 \times 10^4$ ($p = 0.075$) is resolved by the deeper-annealed, higher- n experiment of Fig. 2 ($p = 6.5 \times 10^{-5}$). Only the fixed- N true–shuffle gap is a controlled comparison.

only the rank correlation ρ does, and there the true sequence is significantly *less* sorted. The folded conformation thus behaves as a physical readout sensitive not to the composition of the stiffness sequence but to its sequential arrangement—specifically to correlations that a random permutation destroys.

5.2 Falsification of the local-trap hypothesis and the emergence of macroscopic order

In version 1.0 of this work we advanced a concrete, falsifiable hypothesis for the ordering signature of Sec. 4.3. Observing that dense clusters of small gaps—twin primes (gap 2) alone account for $\approx 14\%$ of all gaps [21]—we conjectured that these clusters act as *local topological traps*, anchoring neighbouring rigid rods within the collapsed interior and thereby *reducing* the stiffness–radius rank correlation ρ below that of an order-randomized shuffle. This hypothesis carries a sharp, testable prediction: the correlation length responsible for the signal must be *short*, so that preserving gap order over modest contiguous windows already suffices to reproduce the true-chain suppression. We have now tested that prediction directly—with the scale-selective surrogates anticipated in version 1—and find it false.

We introduce a complementary pair of surrogates—a “scissors” test—each of which preserves the gap multiset *exactly* (and hence the marginal stiffness distribution of the shuffle S1), differing only in the scale at which sequential order is destroyed. Partition the ordered prime-gap sequence into contiguous blocks of length L (measured in gaps). The *block shuffle* (S4) retains the gap order *within* each block but randomizes the arrangement *of* the blocks, destroying long-range order while preserving local order; it interpolates from S1 at $L = 1$ to the true chain at $L \geq n_{\text{gap}}$. The *local scramble* (S5) performs the converse operation—it holds the blocks in their true global positions

but randomizes the gaps *within* each block, destroying local order while preserving the large-scale arrangement—and interpolates from the true chain at $L = 1$ to S1 at $L \geq n_{\text{gap}}$. Both are evaluated on the common axis $f(L) \equiv (\rho_{\text{S1}} - \rho)/(\rho_{\text{S1}} - \rho_{\text{true}})$, with $f = 0$ shuffle-like and $f = 1$ true-like. All arms use $N = 10^4$, $n = 40$, and the reference annealing protocol of Sec. 2 (Table 1; 2000 steps, 0.20 steps/bead), so that each surrogate is directly comparable to the true and S1 ensembles of Sec. 4.3.

The two blades give opposite and decisive answers (Fig. 5). Destroying *long-range* order abolishes the signal at every scale probed: S4 is statistically indistinguishable from the shuffle for all $L \in \{10, 50, 100, 500\}$ (S4 vs. S1: $p \geq 0.10$, $|d| \leq 0.38$), never descending toward the true chain. Destroying *local* order, by contrast, leaves the signal fully intact: at $L = 50$ and $L = 100$ the local scramble reproduces the true chain to within noise (S5 vs. true: $\rho = -0.025 \pm 0.010$, $p = 0.91$, $d = -0.03$; and $\rho = -0.029 \pm 0.008$, $p = 0.62$, $d = -0.11$), while differing from the shuffle at the *full* strength of the original ordering effect ($d = -1.07$ and $d = -1.37$; $p = 1.3 \times 10^{-5}$ and 5.8×10^{-8})—an effect size matching the $d = -1.05$ of the primary true-vs-shuffle contrast (Sec. 4.3). The imprint thus survives complete randomization of gap order within windows of one hundred consecutive gaps—windows that shred every twin- and cousin-prime cluster in place—yet vanishes the instant the large-scale arrangement of those windows is disturbed.

This constitutes a clean falsification. Local order is neither *sufficient* (S4 retains it and the signal dies) nor *necessary* (S5 destroys it and the signal persists) to reproduce the true-chain suppression. The local topological-trap mechanism conjectured in version 1 is therefore excluded as the driver at $N = 10^4$. The result simultaneously sharpens the decomposition of Sec. 4.3: the Cramér surrogate S3, which retains only the coarse $1/\ln i$ density trend [13], does *not* reproduce the true chain ($\rho_{\text{S3}} = +0.045$, statistically near S1), whereas S5, which retains the true *correlated* arrangement on scales $\gtrsim L$, does. What the fold registers is thus neither the marginal composition of the gap sequence nor its fine local texture, but its genuine long-range arithmetic order.

Because the local-scramble blade is free of the combinatorial starvation that inflates the variance of S4 at large L (at $L = 500$ only three blocks remain, and the block-permutation ensemble collapses to $3! = 6$ arrangements), it further permits a direct estimate of the correlation length. The retention $f_{\text{S5}}(L)$ falls from unity through half-maximal at $L = \xi$, giving

$$\xi \approx 3.5 \times 10^2 \text{ gaps} \quad (68\% \text{ CI } [1.8, 4.9] \times 10^2), \quad (8)$$

a scale spanning $\approx 30\%$ of the $n_{\text{gap}} = 1228$ -gap sequence at $N = 10^4$. We emphasize that ξ , like every observable in this paper, is a property of the *annealed* ensemble and is therefore conditioned on the annealing protocol: Eq. (8) is the half-retention scale under the reference protocol (2000 steps, 0.20 steps/bead), not a protocol-independent equilibrium constant. The order responsible for the topological imprint is, in this precise sense, *macroscopic*: it resides not in the neighbourhood of any individual prime but across a finite fraction of the entire progression sampled. We note that ξ exceeds the few-lag range over which prime-gap pair-clustering is concentrated [22], indicating that the fold responds to correlations beyond nearest-neighbour clustering.

Note added (Version 3.0). Follow-up work by the author (R. Chen, companion manuscript, in preparation), which measures $\xi(N)$ across $N = 10^4$ – 8×10^4 at an annealing depth matched across sizes and deeper than the reference protocol used here, re-measures the $N = 10^4$ crossover under that deeper protocol and obtains $\xi \approx 4.8 \times 10^2$ gaps (68% bootstrap CI $[4.1, 5.7] \times 10^2$; $\xi/n_{\text{gap}} \approx 0.39$). The deeper-annealed value lies at the upper edge of the interval in Eq. (8): deeper annealing lets the chain express the arithmetic ordering out to somewhat larger block scales. The two measurements are protocol-conditioned estimates of the same crossover and are compatible in this sense; both place ξ at a macroscopic fraction (~ 0.3 – 0.4) of the sequence, so the falsification of

the local-trap hypothesis and the macroscopic-order conclusion of this section are unaffected. The protocol sensitivity of the *absolute* value of ξ should nonetheless be borne in mind wherever the number in Eq. (8) is quoted.

Two caveats bound this conclusion. First, ξ is estimated from a four-point retention curve at a single size; the wide interval in Eq. (8) reflects this, and a finer S5 sweep over $L \in [150, 800]$ would tighten it. Second, and more consequential, whether ξ grows in proportion to N —the hallmark of scale-free long-range order—or saturates at a fixed number of gaps—a genuine finite correlation length—cannot be decided from a single N ; resolving this dichotomy is the central question the present result raises. We also record a mild, sub-significant over-suppression by S5 at the smallest window ($L = 10$: $\rho = -0.046 \pm 0.009$, $d = -0.39$ vs. true, $p = 0.089$), consistent either with statistical fluctuation or with a weak counteracting role of the finest local order. Within these bounds the physical picture that replaces the local trap is unambiguous in direction: the true prime sequence stamps a residue of *macroscopic* arithmetic order onto the folded conformation, and it is this long-range structure—not local clustering—that the polymer reads. In the language of version 1’s closing sentence, which held that “ordering matters, not yet how,” we can now state that the *how* is macroscopic.

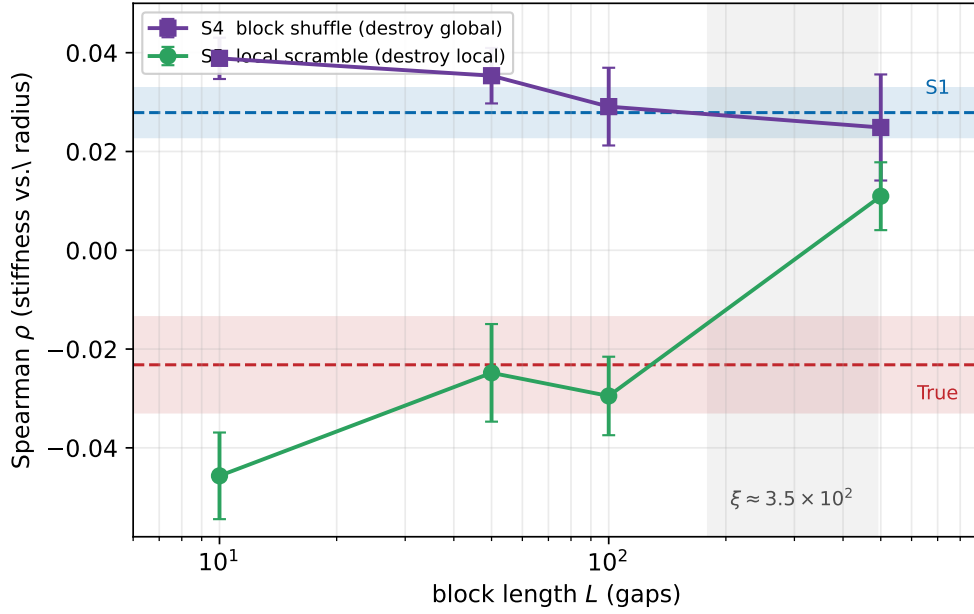


Figure 5: The S4/S5 scissors at $N = 10^4$, $n = 40$ per arm, reference annealing protocol (2000 steps, 0.20 steps/bead). Spearman ρ (stiffness vs. radius) against block length L , with the true-prime and shuffle (S1) reference bands (mean $\pm$ s.e.m.). **S4** (block shuffle; destroy global order, keep local) remains statistically pinned to the shuffle band at every L (vs. S1: $p \geq 0.10$). **S5** (local scramble; destroy local order, keep global) remains pinned to the true-prime band up to $L \approx 10^2$ (vs. true at $L = 50, 100$: $p = 0.91, 0.62$) and departs toward the shuffle only for $L \gtrsim \xi$. The two blades cross near the half-retention scale $\xi \approx 3.5 \times 10^2$ gaps (shaded, 68% CI $[1.8, 4.9] \times 10^2$), the correlation length over which the core-shell ordering collapses.

5.3 Connections and relation to prior work

Our result offers a physical manifestation of the fact that the prime-gap sequence is neither periodic nor memoryless [12, 21, 22]: a folding process, despite the drastic coarse-graining it imposes, still transmits a residue of arithmetic order into three-dimensional structure. In this sense the polymer acts as an unconventional probe of long-range correlations in the primes, complementary to spectral and combinatorial approaches [8, 9] and to statistical analyses reporting $1/f$ -type long-range correlations in the distribution of the primes themselves [23], with which the macroscopic correlation scale found here is qualitatively consonant. The general idea of using polymeric compaction as a readout of long-range correlations in a one-dimensional data sequence was proposed by Odijk [30], who suggested extracting informational correlations from the shape anisotropy of a globule collapsed under a sequence-derived energy; our construction differs in mapping the sequence onto a mechanical (bending) field rather than an interaction energy, in reading out rank statistics of the radial organization rather than shape harmonics, and—most importantly—in the marginal-matched, scale-selective surrogate hierarchy that permits a controlled attribution of the signal to ordering at a measured scale.

Two bodies of prior work frame the polymer-physics side of the result. First, the conformation-dependent sequence-design programme of Khokhlov and Khalatur [26] established the *converse* map to ours: coloring a collapsed homopolymer globule by radial position generates “protein-like” HP sequences, and these globule-derived sequences were shown to carry Lévy-flight-type long-range correlations [27]. That programme runs from structure to sequence and finds that long-range-correlated sequences are precisely those *adapted* to core-shell organization; we run from sequence to structure and find that the primes’ particular long-range order *resists* the radial stiffness sorting that marginal-matched random orderings undergo. The two results are complementary rather than contradictory—the designed correlations of Ref. [27] are constructed to be commensurate with a radially sorted globule, whereas the arithmetic correlations of the primes are quenched constraints with no such adaptation—and together they indicate that the *sign* of a long-range-correlation effect on radial organization depends on the structure of the correlations, not merely on their presence. Second, mean-field analyses of random heteropolymers with power-law-correlated sequences [28] and studies of correlated-sequence copolymer ensembles found that sequence correlations reshape freezing and adsorption behaviour; our contribution relative to that line is the single-sequence, surrogate-controlled setting, in which the correlations are deterministic and the attribution is made against an exactly marginal-matched null rather than across random ensembles. Prior work mapping a deterministic aperiodic sequence onto a molecular chain—Fibonacci-arranged DNA segments [29]—examined conformational scaling of the aperiodic chain itself; the ordering-versus-composition question addressed here, which requires the surrogate machinery, was not posed there.

Finally, the macroscopic correlation length identified in Sec. 5.2 invites a number-theoretic interpretation in light of the effective limit-periodicity and hyperuniformity of the primes [24, 25]: it is natural to conjecture that the fold registers a residue of precisely this multiscale order. Because hyperuniformity is a statement about the *two-point* structure (the small- k structure factor), the spectrally matched surrogate proposed in Sec. 5.4 would test this conjecture directly: if an amplitude-adjusted, spectrum-preserving randomization reproduces the true-chain suppression, the fold reads the two-point (limit-periodic/hyperuniform) layer of the primes’ order; if it does not, higher-order arithmetic structure is implicated. On the physical side, the construction sits within the study of aperiodic and deterministic order—quasicrystals, aperiodic tilings, and sequence-programmed matter [10, 11, 14]. We are, however, cautious about extrapolation: the effect is small in absolute terms, and we make no claim of a route to functional metamaterials. What we demonstrate is narrower and cleaner: that a deterministic, aperiodic arithmetic sequence leaves a detectable, reproducible imprint on the

equilibrium-like conformation of a physical model, distinguishable from every marginal-matched null.

5.4 Limitations

Several limitations bound these conclusions. First, the effect is quantitatively small: local stiffness accounts for only $\sim 1\%$ of the radial-rank variance, the distributions overlap heavily, and no single conformation can be classified as “prime” or “shuffled”—the signal is a shift in the ensemble mean. Second, our ensembles are generated by simulated annealing rather than sampled from a true equilibrium distribution; absolute observables depend on the protocol, and only the fixed- N true-surrogate contrast, annealed identically, is controlled. Third, we have examined a single stiffness map—one α , one functional form, one confinement; robustness across this parameter space remains to be charted. Fourth, our sizes reach only $N = 4 \times 10^4$; the absolute correlation drifts upward with N , and while the gap is significant at every size once statistics and annealing depth are controlled, its fate in the thermodynamic limit is genuinely open. Fifth, as made explicit in Sec. 5.2, the correlation length ξ is conditioned on the annealing protocol: its absolute value shifts with annealing depth (see the note added in Sec. 5.2), and only its order of magnitude—macroscopic, a substantial fraction of the sequence—is protocol-robust within the depths explored. Sixth, two inexpensive robustness controls remain to be run and are natural next steps: (i) a *partial* rank correlation between stiffness and radius controlling for backbone index i (the $1/\ln i$ density decay makes the mean stiffness drift monotonically along the chain, so chain-end effects could in principle contribute to ρ), together with a sequence-reversal control; and (ii) a spectrally matched surrogate (e.g. amplitude-adjusted Fourier transform), which preserves the two-point autocorrelation of the gap sequence while destroying higher-order structure and would determine whether the fold reads only second-order statistics. Finally, the signal is carried by the sensitive rank estimator ρ and is not resolved by the tail-based ΔR at the largest size. We emphasize that this is a controlled in-silico construction: we make no claim that prime numbers are physically privileged in nature, and none of biological relevance.

6 Conclusion

We introduced the PDTP model and used it to ask whether the folded state of an aperiodic-but-deterministic sequence is determined by its composition alone or also by its arithmetic ordering. By comparing the true prime chain against surrogates that preserve the marginal stiffness distribution exactly while destroying order, we found the answer to be affirmative but bounded. The gross conformation is surrogate-invariant—neither R_g nor ΔR distinguishes true from shuffle—so overall size and coarse morphology are fixed by the marginal distribution. The finer organization is not: the stiffness-radius rank correlation is significantly suppressed for the true sequence ($\rho = +0.105$ vs. $+0.137$ at $N = 4 \times 10^4$; $p = 6.5 \times 10^{-5}$, $d = -0.96$), an effect a three-way decomposition attributes to arithmetic ordering rather than to the gap distribution, and reproducible across sizes once statistics and annealing depth are controlled. The effect is quantitatively modest ($\sim 1\%$ of the radial-rank variance) and is a property of the ensemble rather than of any single fold. Version 2.0 resolves the mechanism left open in version 1. Two scale-selective surrogates act as a *scissors* on the correlation length: a block shuffle (S4), which destroys long-range order while preserving local order, and its converse, a local scramble (S5), which preserves the global arrangement while destroying local order (Sec. 5.2, Fig. 5). The two blades give opposite and decisive verdicts. Destroying *local* order leaves the ordering signature fully intact—S5 is statistically indistinguishable from the true chain for windows up to $\sim 10^2$ gaps ($p = 0.62\text{--}0.91$)—whereas destroying *global* order abolishes it, S4 remaining indistinguishable from the shuffle at every scale ($p \geq 0.10$). Local order is therefore neither

sufficient nor necessary to reproduce the signal, and the local-cluster “topological-trap” hypothesis is ruled out: the imprint is driven not by twin-prime clustering but by an irreducible, *macroscopic* long-range arithmetic correlation of length $\xi \approx 3.5 \times 10^2$ gaps under the reference protocol—a scale spanning $\approx 30\%$ of the sequence, and shifting to ≈ 0.39 of the sequence under deeper annealing (note added, Sec. 5.2)—so that the macroscopic character of the scale, rather than its precise value, is the protocol-robust statement. The correlation scale responsible for the signal, identified as the most immediate open question in version 1, is thus now measured up to this protocol conditioning. Beyond the specific case of primes, the PDTP framework and its surrogate-null methodology offer a controlled testbed for how one-dimensional deterministic order is—and is not—expressed in three-dimensional structure. The remaining frontier is the scaling of ξ with system size—whether it grows in proportion to N , the signature of scale-free long-range order, or saturates at a fixed gap count, a genuine finite correlation length—together with the robustness of the effect across the stiffness-map parameter space.

Data and code availability

All code, data, and figure-generation scripts that reproduce the results of this paper are openly available at <https://github.com/Ruqing1963/prime-driven-topological-polymer>. The repository contains the pre-flight calibration, the surrogate/null-model decomposition, the finite-size sweep, the decisive $N = 4 \times 10^4$ experiment, and the conformation generator, together with the ensemble result files and a script that regenerates all figures. Version 2.0 additionally provides the two scale-selective surrogate drivers—the block shuffle (`block_shuffle_test.py`, S4) and the local scramble (`block_local_scramble_test.py`, S5)—with their $N = 10^4$ ensembles and the scissors figure. For long-term preservation, this release (version 3.0) is archived on Zenodo under DOI 10.5281/zenodo.22020688; the version 3.0 changes relative to version 2.0 are enumerated in the Version 3.0 note above and in the repository `CHANGELOG`.

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